MULTI-AGENT REINFORCEMENT LEARNING · RESEARCH REPORT

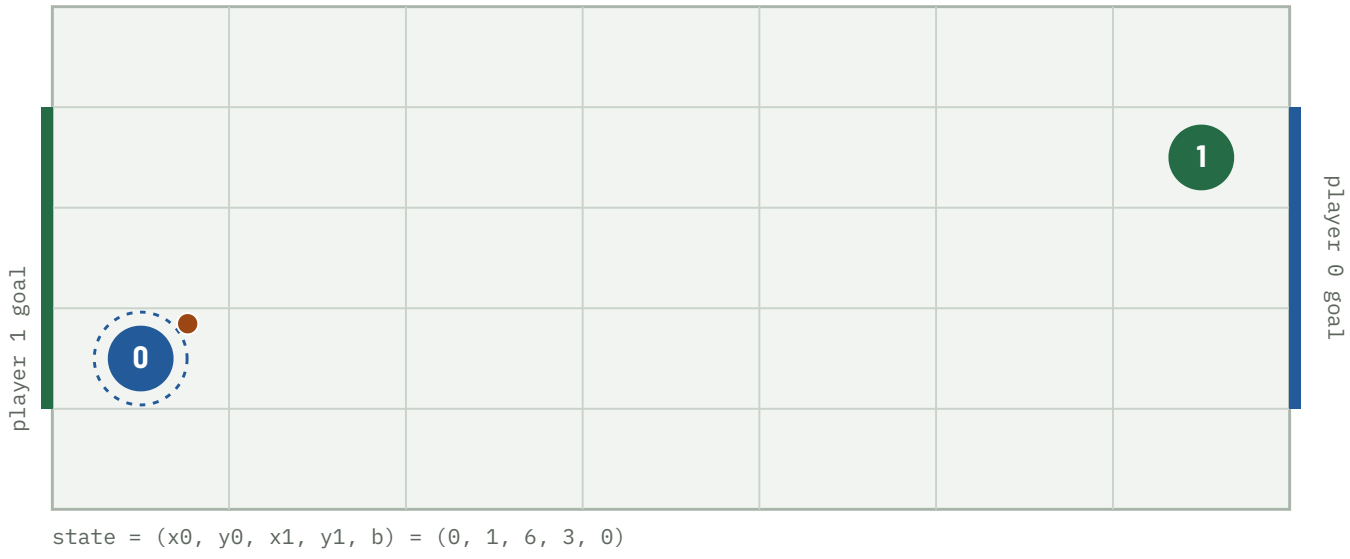
When Does Soccer Need Mixed Strategies?

A pure-first Nash-Q approach: where a two-player soccer Markov game does — and does not — need mixing, and how much LP work that saves.

FOUR QUESTIONS

RQ1 Existence — under what transition structures does a pure equilibrium exist at every state? **RQ2 Efficiency** — how much work does checking for a pure saddle first eliminate? **RQ3 Mechanism** — which spatial configurations force mixing? **RQ4 Robustness** — how do discounting and tolerance affect the split?

repo · soccer-markov-nash | 210+ tests | game family in docs/design.md | methods in docs/methods.md | every policy drawn in docs/gallery.html



The kickoff. 7×5 grid, 2 380 non-terminal states. Player 0 (blue) carries the ball and attacks the right edge; player 1 (green) the left. Start positions are the ones the A10 page assigns this netID. Actions U D L R are chosen simultaneously; reward is +1 / −1 / 0, zero-sum.

ABSTRACT

1 **The deterministic A10 game — solved exactly, undiscounted, by 100-step backward induction — has a pure saddle at every one of its 238 000 (state × step) stage games.** A pure equilibrium exists; the LP is never called; discounting was never load-bearing.

2 **A coin-flip tie-break does not change this;** Littman's random *move order* does — but only when the goal mouth is more than one cell wide, and even then on 3.95% of states (this configuration).

3 **Those mixed states are geometrically localised:** a decision tree predicts them from the state with precision 0.96, and 68 of 94 have a 2×2 matching-pennies support.

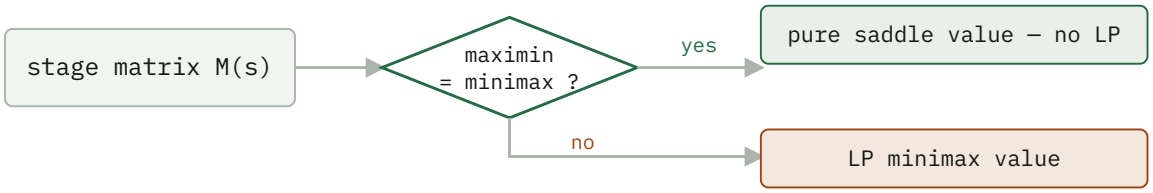
4 **The pure-first hybrid solver** matches the all-LP value function to 4e−16 while calling the LP on only 3.95% of states — ~25× fewer per sweep; mirror symmetry halves the work again.

Method

The game is zero-sum, so at every state the stage game is a payoff matrix $M(s)[a_0, a_1] = E[R + \gamma V(s')]$ and the Shapley (1953) fixed point is the minimax value:

$$V(s) = \mathrm{val}(M(s)) = \max_{p} \min_{k \in 3} (p^{\top} M)[a_1] = \min_{p} \max_{a_0} (Mq)[a_0]$$

The $Q = R + \beta\,\mathrm{Nash}(Q')$ update is this operator. The A10 game is *undiscounted* (±1 at a goal, tie after 100 steps); $\gamma < 1$ is a contraction device for value iteration and RQ4 shows the split holds across γ. Three stage backups: **pure** (the maximin security value — a fast lower bound, not a Nash value when it is below the minimax), **mixed** (the LP minimax value, via `scipy` HiGHS), and **hybrid** (the pure saddle value where `maximin = minimax`, the LP elsewhere — exact everywhere). Support enumeration (`support_enum.py`) finds *all* equilibria, including of general-sum games.



The pure-first hybrid. On the deterministic game the "yes" branch is taken at every one of the 2 380 states, so the LP is never called.

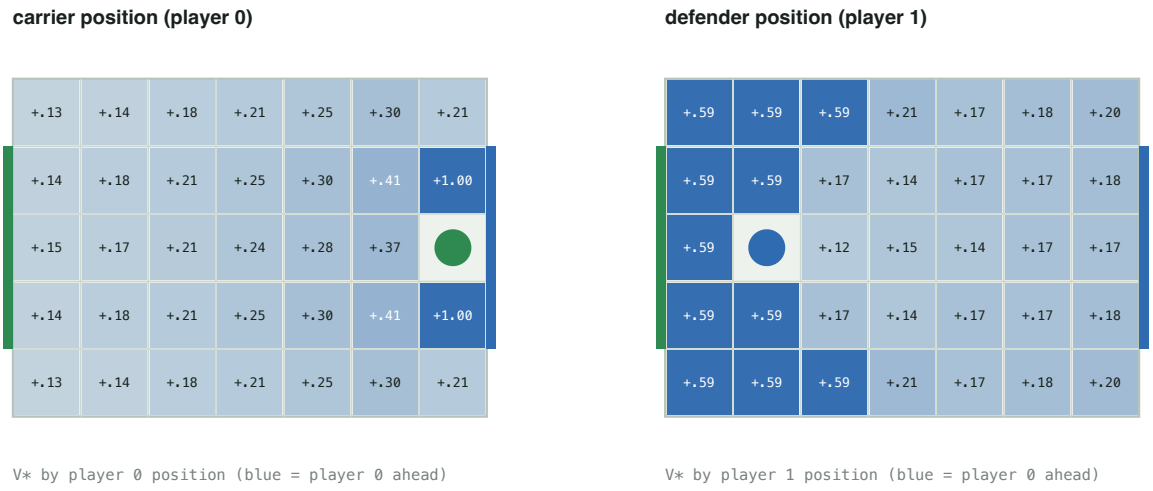
RQ1 — Existence

The A10 game is undiscounted, so it is solved exactly by 100-step backward induction (`run_finite_horizon`, `experiments/undiscounted.csv`):

EXACT UNDISCOUNTED GAME — 238 000 (STATE × STEP) STAGE GAMES

MOVE ORDER	MIXED STAGE GAMES	V(KICKOFF)	PURE EQUILIBRIUM?
deterministic (exact A10)	0	0.000	yes — pure (non-stationary)
coinflip tie-break	0	0.000	yes
random move order	3 148 (404 states)	+0.459	no

The deterministic equilibrium is **constructive and memoryless**: each player's win attractor (states from which it forces a goal, `soccer_nash/attractor.py`) equals its $V^* = \pm 1$ set, and the pure profile "attractor move on your winning set, safety move elsewhere" realizes V^* at every state (`scripts/positional.py`) — the game is positionally determined.



V* by player 0 position (blue = player 0 ahead)

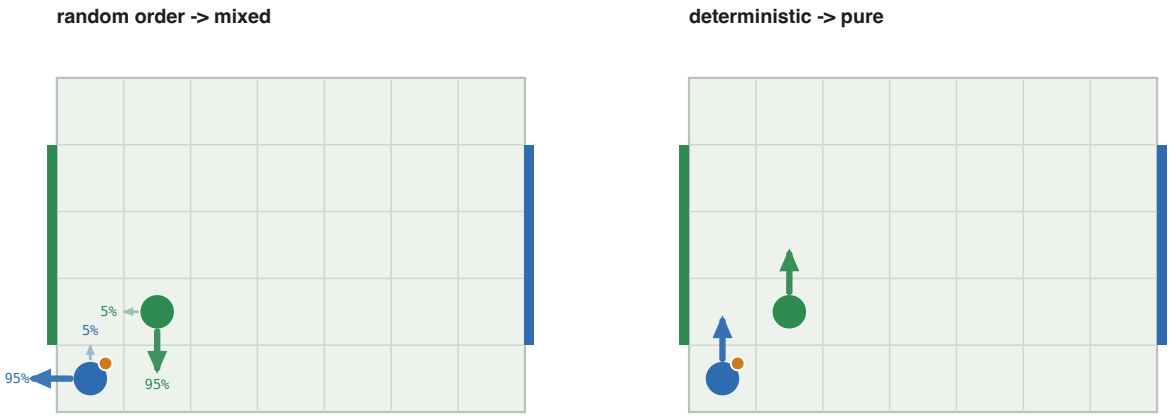
V* by player 1 position (blue = player 0 ahead)

The value surface of the random-order game ($\gamma = 0.9$). Left: V^* by carrier position — a smooth gradient toward the attacking goal, saturating to +1 on the two cells that score in one move. Right: V^* by defender position with the carrier held near midfield — nearly flat, the defender's location barely moves the value.

The stationary $\gamma < 1$ solve agrees and is faster — 0 no-pure-saddle stage games on the deterministic game at every γ from 0.5 to 0.995, and **94** / **2380** on the random game at $\gamma = 0.9$ (`V(kickoff) = +0.164` at the netID start). The no-pure-saddle count never depends on the kickoff.

IT IS THE MOVE ORDER, NOT THE RANDOMNESS

The deterministic and coin-flip value functions are *identical* (`max |V_det − V_coin| = 0` at every γ): optimal play never enters a losing contest, so the coin is never flipped. Only Littman's random move order — where a ball-steal depends on who resolves first *and* on both players' targets — creates matching-pennies subgames.



mixed equilibrium · V = +0.078

pure equilibrium · V = +0.000

One state, two resolution rules. Arrows are the equilibrium action distribution, sized by probability. Random order (Littman) makes the stage game matching pennies; deterministic carrier-wins collapses it to a pure best reply. Full set: `docs/gallery.html`.

RQ2 — Efficiency

Two numbers, kept apart: the **LP-call rate** is a property of the game and reproduces exactly; **wall clock** is a median of 5 repeats and depends on the machine and LP backend (`scripts/benchmark.py`).

RANDOM MOVE ORDER, Γ = 0.9 — THE CASE WHERE THE LP IS ACTUALLY NEEDED

SOLVER	SWEEPS	LP CALLS	STATES NEEDING LP	MEDIAN WALL CLOCK
pure (lower bound)	18	0	—	0.3 s
hybrid exact	108	9 672	3.95%	13.2 s
all-LP exact	108	~3.6 M	100%	~8 min

The hybrid solves an LP on only the **3.95%** of states lacking a pure saddle — a ~25× per-sweep reduction — and matches the all-LP value to 4e−16. The wall-clock ratio (~13 s vs ~8 min) is larger but less portable: it also reflects matrix construction and the LP backend. **Mirror symmetry** (`run_symmetric`, deterministic dynamics only): every state has a distinct board-flip-plus-player-swap mirror, so the 2 380 states are 1 190 pairs — reconstruct one from the other by `V(mirror(s)) = −V(s)`, cutting the deterministic solve 0.33 s → 0.23 s.

Freeze-then-iterate, and its staleness

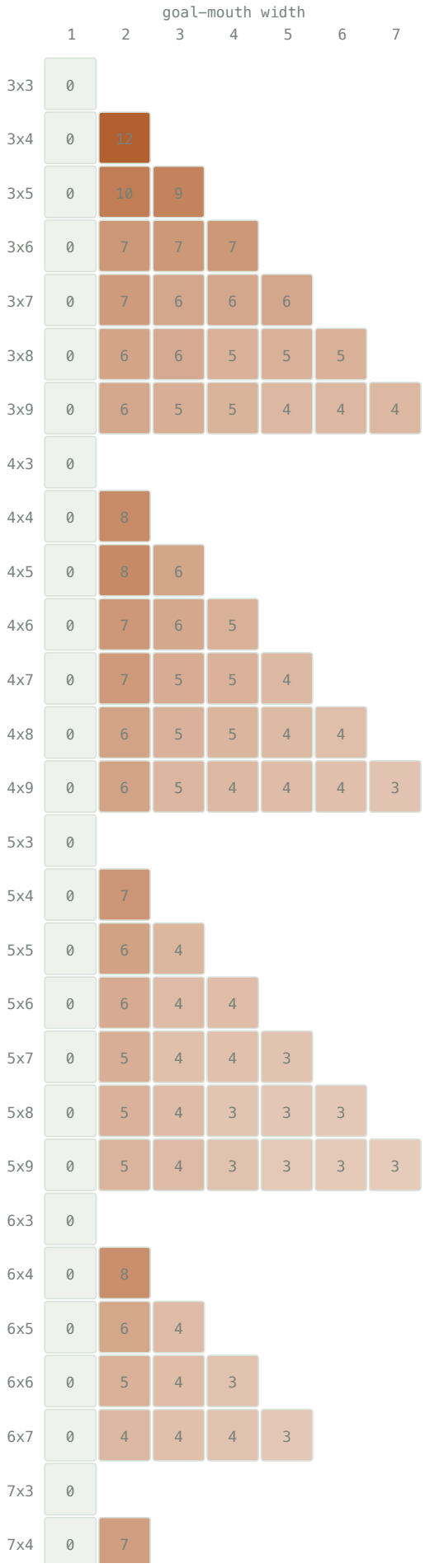
RANDOM MOVE ORDER, HYBRID SOLVER — SAME FIXED POINT, |DIFF| < 3E−9

	ROUNDS / SWEEPS	MATRIX-GAME SOLVES	WALL CLOCK
value iteration	108	9 672	13.8 s
policy iteration	21	1 879	17.2 s

5× fewer LP solves — but the frozen strategies' distance to the current Nash stays *maximal* (a full pure-strategy flip) for **16 outer rounds**, then collapses to < 1e−8. The thrashing eats the saving; on the deterministic game it is strictly worse. Value iteration with the per-sweep Nash cache wins here.

RQ3 — Mechanism

Goal-mouth width, not board size, is the gate. The phase diagram sweeps every board with `width ≤ 11`, `height ≤ 9`, `width·height ≤ 45` against goal widths 1 to `height−2` (`scripts/phase_diagram.py --analyze`, `experiments/phase_diagram.csv`).


goal–mouth width
1 2 3 4 5 6 7
3x3 0
3x4 0 12
3x5 0 10 9
3x6 0 7 7 7
3x7 0 7 6 6 6
3x8 0 6 6 5 5 5
3x9 0 6 5 5 4 4 4
4x3 0
4x4 0 8
4x5 0 8 6
4x6 0 7 6 5
4x7 0 7 5 5 4
4x8 0 6 5 5 4 4
4x9 0 6 5 4 4 4 3
5x3 0
5x4 0 7
5x5 0 6 4
5x6 0 6 4 4
5x7 0 5 4 4 3
5x8 0 5 4 3 3 3
5x9 0 5 4 3 3 3 3
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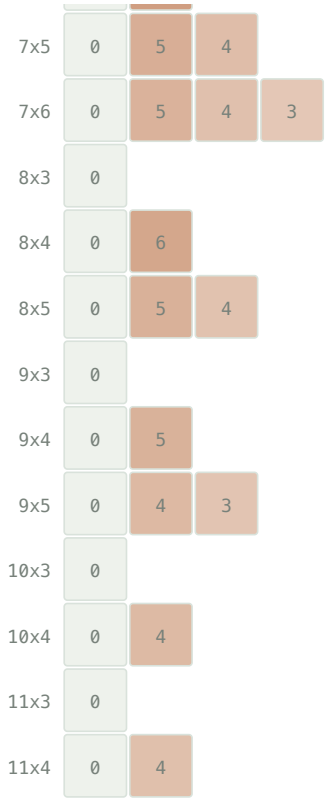
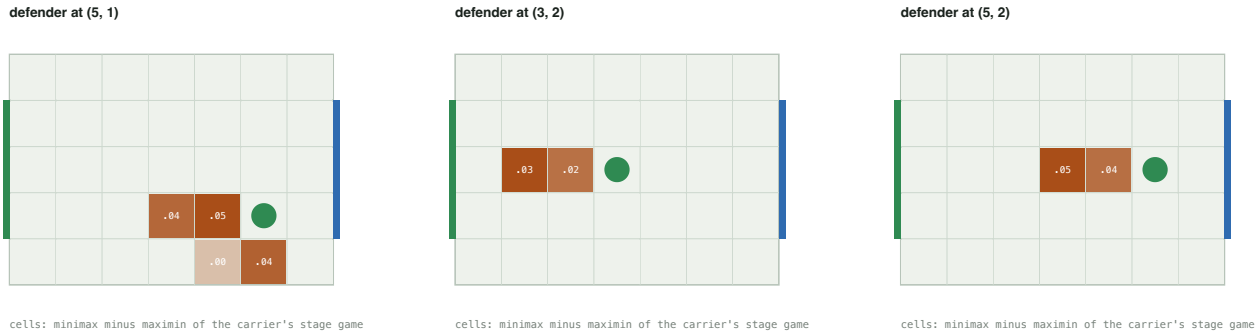
6x7 0 4 4 4 3
7x3 0
7x4 0 7



The gate. Goal width 1 → **0** mixed states, on all 40 one-cell configs. Goal width ≥ 2 → mixing on all 87: whether a board mixes is a *perfect* function of goal width. The *fraction* (2.7–12%) is an OLS R^2 0.64 in board area — a dilution effect — with goal width adding little and a negative coefficient. Every state is reachable from the kickoff.



Where the carrier has to guess. Defender pinned at the green disc; each cell shaded by `minimax − maximin` of the carrier's stage game (0 = pure saddle). Mixing is a local phenomenon — a handful of cells where the defender contests the forward lane.

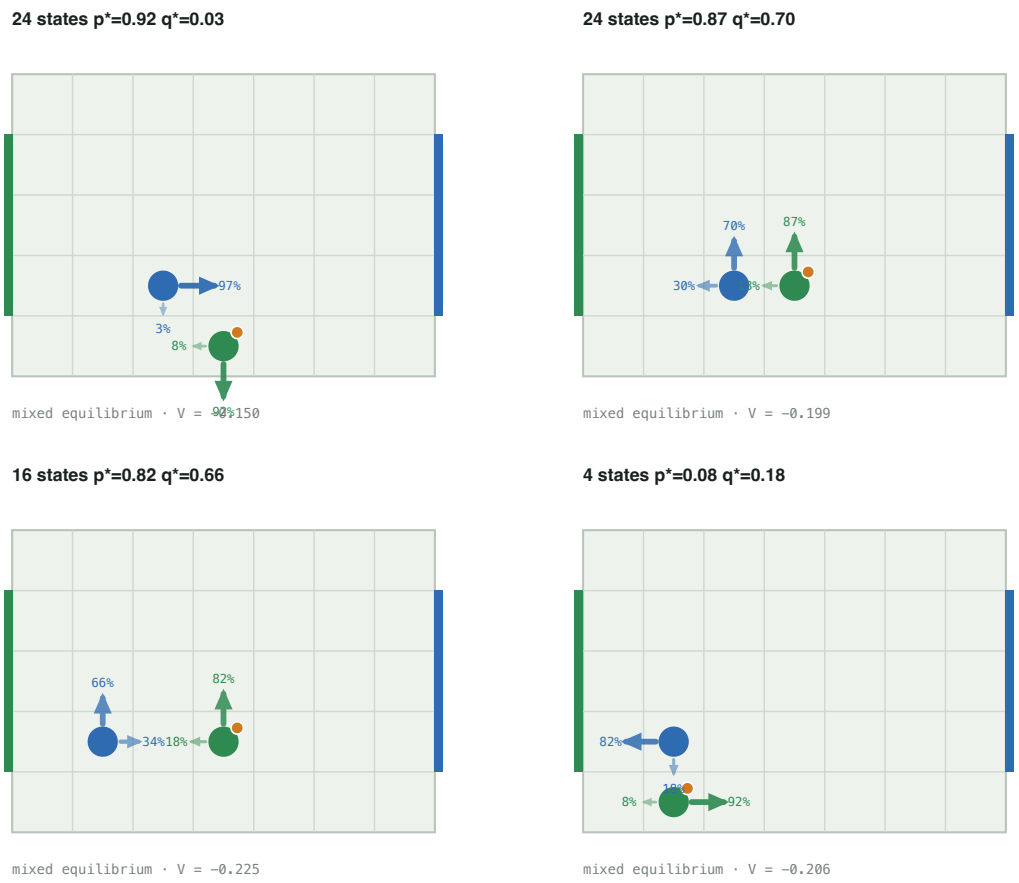
Geometry predicts the label (`scripts/geometry_model.py`): a depth-4 decision tree on carrier-frame features separates `mixed` from the rest with **precision 0.96, recall 0.91**. The dominant feature is `defender_can_intercept` — the defender within one move of the carrier's forward cell (`P(mixed) = 0.44` vs `0.002`).

Beyond prediction, the 94 mixed states canonicalize under the board mirror to 47 pairs and cluster into **8 geometric templates** (`scripts/templates.py`, `docs/templates.md`).

THE MIXED-STATE TEMPLATES — CARRIER-FRAME GEOMETRY, EQUILIBRIUM SUPPORT

STATES	GEOMETRY	SUPPORT	STRUCTURE
24	defender 1 ahead, carrier one row off the goal row	2×2	matching pennies
24	defender 1 ahead, same row, carrier on a goal row	2×2	matching pennies
16	defender 2 ahead, same row, carrier on a goal row	2×2	matching pennies
4	defender diagonally adjacent, carrier off the goal row	2×2	matching pennies
14	defender 1–2 ahead, same row, on a goal row	2×1	near-pure saddle
8	defender 2 ahead, same row, on a goal row	3×2	degenerate (row ties)
4	defender 2 ahead, carrier off the goal row	3×3	wider mix

The **four 2×2-support templates are all verified matching pennies** — the carrier's best reply flips between the two defender columns and vice versa — covering **68 of 94** states: carrier {climb, advance} against defender {cover a lane, hold the forward cell}. All 94 share one value across every equilibrium; 64 have a unique one.



One policy fan per matching-pennies template. Each covers a class of mixed states related by carrier-frame geometry; p*, q* are the equilibrium weights on the first support action. The carrier randomizes between two scoring lanes, the defender between covering them.

Why the mix is unavoidable. A mixed stage game needs all three of: a stochastic (½–½) resolution order; a goal mouth wider than one cell, so the carrier has two scoring lanes; and a defender close enough to contest the forward cell but unable to cover both lanes in one move. The carrier climbs or drops, the defender guesses which — matching pennies. Drop any one clause — a single goal cell, deterministic resolution, or the coin-flip tie-break — and every stage game has a pure saddle. The full argument and one stage matrix per template are in `docs/templates.md`. For the single goal cell the theorem is partly proved (`docs/proof.md`): the defender has a closed-form optimal strategy — guard the one cell — and every stage game is dominance-solvable, machine-checked for all boards up to 11×5.

SHALLOW MIXES, BUT THEY COMPOUND

Every one of the 94 no-pure-saddle stage games sits within `0.07` of a pure saddle (`minimax − maximin`, median `0.029`) — the equilibria are shallow, and the exact mixing weights are numerically delicate. What is robust: 68 have a 2×2 support and all 68 are structurally matching pennies, and the *value of mixing* — `V(hybrid) − V(pure maximin)` — is `+0.13` on average at those states. It compounds along a path: at the centred kickoff a pure-strategy player secures only a draw (`0.000`) where mixing is worth `+0.15` (`scripts/mixing.py`, `docs/mixing.md`).

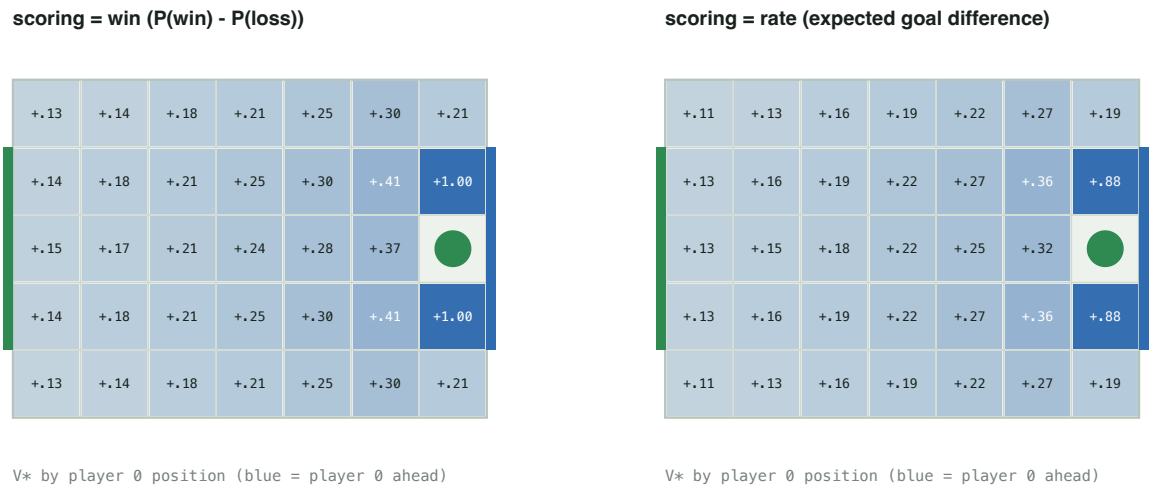
The reward objective does not move the mixed region

The default game rewards a *win* — first goal ends it, value `P(win) − P(loss)`. A second objective, `scoring="rate"` (`scripts/reward.py`), keeps playing: a goal scores `±1` and restarts with the ball to the conceding team, so the value is the expected discounted *goal difference* and `γ < 1` is load-bearing. On the 7×5 random-order board:

SAME DYNAMICS, TWO REWARD OBJECTIVES — RANDOM MOVE ORDER, DISCOUNT 0.9

OBJECTIVE	NO-SADDLE STATES	VALUE RANGE	V(KICKOFF)
`win — P(win) − P(loss)`	94	`[−1.00, +1.00]`	`+0.150`
`rate — expected goal difference`	`94 — the same 94`	`[−0.88, +0.88]`	`+0.132`

The no-pure-saddle set is *identical* — **0** states enter or leave. A mixed Nash equilibrium is an equilibrium of a single stage game `M(s)`; changing how the horizon is scored rescales `M(s)` through `V(s′)` but does not cross the `maximin = minimax` boundary. The value *range* compresses (`±1 → ±0.88`): under `rate` a lost position is not a cliff — conceding restarts play with possession, a floor under every state. Details in `docs/reward.md`.



Same shape, compressed range. `V*` by carrier position under the two objectives. `rate` pulls the goal-adjacent `+1` cells down to `+0.88` — the concede-and-restart floor — without changing where the value gradient runs.

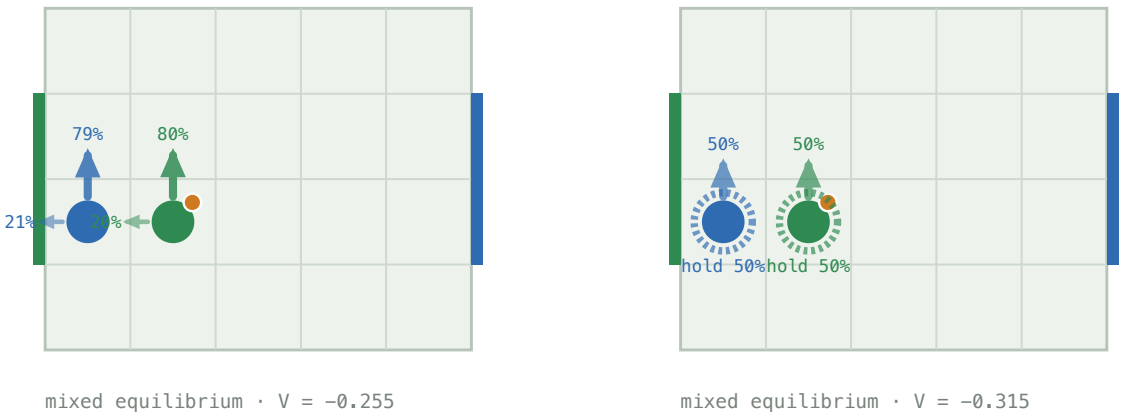
Littman's fifth action

Littman's 1994 soccer game has a fifth action, `stand`, and his Figure 2 — a carrier pinned against its own goal by an adjacent defender — is the textbook "soccer needs mixing" example: the carrier must randomize between standing and moving. Adding `stand` to the random-order game (`n_actions=5`, `scripts/littman.py`) on Littman's 5×4 board:

LITTMAN'S BOARD (5×4, 2-CELL GOALS), RANDOM MOVE ORDER, DISCOUNT 0.9

ACTION SET	NO-SADDLE STATES	STAND IN THE MIXED SUPPORT	V(KICKOFF)
`N, S, E, W`	56	0	+0.147
`N, S, E, W, stand`	56	48	+0.172

The fifth action does *not* move where mixing happens — 48 of the 56 no-pure-saddle states are shared, 8 swap in and 8 out at the margin — but it changes *what* the mix is: the carrier now hedges with `stand` in 48 of 56 states rather than with `retreat`, and every player is slightly better for the option (`V(kickoff) +0.147 → +0.172`). The Figure 2 state is a clean matching-pennies matrix with equilibrium (`½ climb, ½ hold`). The deterministic game keeps a pure saddle at every state with five actions too.



Littman's Figure 2. Same state, four moves (left) vs. five (right). The dashed ring is `stand`. Adding it turns the carrier's climb/retreat hedge into climb/hold — Littman's mixed policy.

RQ4 — Numerical robustness

The value is three numbers

`value_bracket` returns what the row player guarantees ($\min_k (pM)_k$), gets ($p^\top Mq$), and can reach ($\max_i (Mq)_i$). The true value is bracketed, so the width certifies the LP error.

FINDING

With `scipy`'s HiGHS the three agree to `1.1e-16` per stage game, `8.7e-10` accumulated. The concern is real for older vertex/simplex solvers — a non-issue here.

Discounting and tolerance

	DETERMINISTIC	RANDOM (7×5)
mixed states, γ = 0.5 → 0.9 → 0.995	0 → 0 → 0	122 → 94 → 102
classification split, rel_tol 1e-12 → 1e-3	stable	604 / 1682 / 94 (stable)
classification, rel_tol 1e-2 / 1e-1	—	80 / 0 mixed
fixed 0.1-rounding flips a saddle	0	62

Discounting mildly shifts which states mix; the classification is otherwise robust across nine orders of magnitude of tolerance, breaking only as the tolerance approaches the real `minimax − maximin` gaps. Fixed 0.1-rounding — proposed to force indifference — misclassifies 62 states, exactly the small-entry mixed region.

Secondary validation — the policy plays correctly

- Duality gap `V0_br(s₀) + V1_br(s₀) < 1e−9` for every move order — no opponent beats the game value.
- Nash vs. Nash reproduces the value: forced draws in the deterministic and coin-flip games; `+0.162 ± 0.002` empirical discounted return over 5 seeds of 3000 games (vs. `+0.164` computed) in the random game.
- A best response to one assumed opponent is fragile: the Part 2 policy has exploitability `0.43` — worse than random — which is the A10 competition's warning about non-equilibrium submissions.
- A10 networks over 5 seeds (`experiments/a10_competition_seeds.csv`): the Part 2 imitation net plays optimally at `0.990 ± 0.000` of states and wins Q8 every time; competition Network Second is exact, but Network First sits at a `0.19 ± 0.02` exploitability from this kickoff that does not wash out with re-seeding.

Beyond zero-sum, and open questions

- **General-sum — extension point.** `markov_game.py` solves 2-player general-sum Markov games by enumerating *all* stage equilibria (`support_enum.py`) and selecting one — the notes' "largest sum of values", a no-op for zero-sum but decisive for Battle of the Sexes, where it avoids the mixed equilibrium whose value is below either pure one. Pure-first throughout. The soccer game is zero-sum, so this does not touch the main result.
- **Function approximation — partial.** A network fit to the stage matrices (`nash_dqn.py`, 5→96→96→16 with biases, 600 epochs) trails the exact solver, over **5 seeds** (`experiments/nash_dqn_seeds.csv`):

	EXACT HYBRID NASH-Q	NEURAL NASH-Q (MEAN ± SD)
max |V − V_exact|	0	0.55 ± 0.02
mean |V − V_exact|	0	0.13 ± 0.00
action agreement	100%	43% ± 2%
pure/mixed classification	100%	67% ± 2%
exploitability	<1e−9	0.44 ± 0.05
convergence (epochs to plateau, of 600)	exact	469 ± 31; 3/5 still creeping
runtime	9 sweeps, 0.3 s	600 epochs, ~35 s

Deterministic game only — `train_nash_dqn` rejects the stochastic variants, so the network never has to represent a genuinely mixed stage game. Even so it trails: it fits the ~1000 zero-value states easily (small *mean* error) but misses the γ^k bands and contested regions — worst-state error `0.55`, policy about as exploitable as random play, mis-calls *whether* a state needs mixing a third of the time. Keep the exact solver as ground truth.

- **Four research questions** — the single-cell theorem's last step, whether the goal-width dichotomy is a known result, the move-order-vs-randomness distinction, and why freeze-then-iterate thrashes — are stated and answered as far as the evidence allows in `docs/discussion.md`.

Limitations

- The A10 page redacts the ID-specific start position and goal rows; this repo uses the standard Littman geometry (configurable). Across the phase-diagram sweep every one-cell-goal board has 0 mixed states and every wider-goal board has some; the exact `3.95%` is the 7×5 / 3-cell-goal case.
- Several collision sub-cases (carrier vs. stationary opponent, bump = steal, swap possession) are *interpreted* from the two stated A10 rules, not quoted. If course staff intended otherwise, Claim A must be re-measured. See `docs/assumptions.md`.
- **Claim A/B** (a pure equilibrium of the deterministic game) is established constructively — the explicit pure memoryless attractor/safety profile. **Claim C** (the theoretical game necessarily has one, over all rules and settings) is *not*, for the general goal mouth; for the single goal cell it is partly proved (`docs/proof.md`).
- `random` and `coinflip` are different game definitions, isolating the collision rule and the tie-break respectively — not the A10 game.

```
soccer_nash — game.py · matrix_games.py · support_enum.py · nash_q.py · markov_game.py ·
numerics.py · dominance.py · attractor.py · geometry.py · symmetry.py · tree.py ·
templates.py · onecell.py · reachability.py · best_response.py · exploit.py · mlp.py ·
nash_dqn.py · opponents.py · shaping.py · simulate.py · experiment.py · render.py · viz.py
scripts — experiments.py · analyze.py · numerics.py · equilibria.py · geometry_model.py ·
templates.py · phase_diagram.py · onecell_proof.py · positional.py · benchmark.py ·
policy_iteration.py · selfplay.py · nash_dqn.py · render.py · gallery.py · littman.py ·
mixing.py · reward.py · a10_part1.py · a10_part2.py · a10_competition.py
docs — design.md · methods.md · discussion.md · assumptions.md · templates.md · proof.md ·
geometry.md · mechanism.md · littman.md · mixing.md · reward.md · gallery.html
experiments — baseline.csv · undiscounted.csv · gamma_sweep.csv · tolerance_sweep.csv ·
board_sweep.csv · goal_mouth_sweep.csv · one_cell_goal.csv · phase_diagram.csv ·
nash_dqn_seeds.csv
env: 7×5 grid, 2 380 states, zero-sum ±1, γ = 0.9
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